

Dormitory Management System API Security Risk Analysis Report

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1. Introduction

This project is a Dormitory Management System with API Security Risk Analysis. The system manages student dormitory information, room allocation, and secure admin access.

2. Objectives

- Analyze API security risks in the dormitory management system.
- Protect student information and admin accounts.
- Apply password encryption and secure authentication.

3. Technologies Used

- HTML
- CSS
- JavaScript
- PHP
- XAMPP
- MySQL Database

4. System Features

- Student registration
- Room allocation management
- Admin login system
- Database management
- Security analysis

5. Security Analysis

The system checks common API security risks such as weak authentication, insecure data exposure, missing input validation, and insecure login mechanisms.

6. Password Encryption

Admin passwords are encrypted before storing in the MySQL database using PHP password hashing functions.

7. Database Security

The project uses XAMPP MySQL database to securely store student information, room data, and login credentials.

8. Recommendations

- Use HTTPS communication
- Apply strong password policies
- Validate all user inputs
- Use secure authentication methods

9. Conclusion

This project demonstrates practical cybersecurity concepts in a real-world dormitory management system focusing on API security and database protection.